

Assignment-2

1. Write a Python program to **iterate over a list and print all its elements**.
2. Write a Python program to **find the sum of all elements in a given list**.
3. Write a Python program to **find the maximum and minimum elements in a list**.
4. Write a Python program to **iterate through a string and count the number of vowels present in it**.
5. Write a Python program to **traverse a tuple and count how many elements are even numbers**.
6. Write a Python program to **create a list of squares of numbers from 1 to 10 using list comprehension**.
7. Write a Python program to **create a new list containing only even numbers from a given list**.
8. Write a Python program to **remove duplicate elements from a list without using the set() function**.
9. Write a Python program to **reverse a list using iteration without using built-in functions**.
10. Write a Python program to **count the frequency of each element in a list**.
11. Write a Python program to **create a set and iterate through it to print all unique elements**.
12. Write a Python program to **perform union, intersection, and difference operations on two sets**.
13. Write a Python program to **iterate through a dictionary and print all keys and their corresponding values**.
14. Write a Python program to **swap keys and values in a dictionary**.
15. Write a Python program to **count the occurrence of each word in a given sentence using a dictionary**.
16. Write a Python program to **print numbers from 1 to 20 along with their squares using the range() function**.
17. Write a Python program to **print elements of a list along with their indices using the enumerate() function**.
18. Write a Python program to **combine two lists into a dictionary using the zip() function**.
19. Write a Python program to **create a custom iterator that generates numbers from 1 to N**.
20. Write a Python program to **implement a generator function to generate Fibonacci numbers up to N**.

21. Write a Python program to **find two numbers in a list that add up to a given target value (Two Sum problem)**.
22. Write a Python program to **find the longest substring in a string without repeating characters**.
23. Write a Python program to **group a list of strings into anagrams** (e.g., ["eat", "tea", "tan"]).
24. Write a Python program to **find the top K frequent elements in a list**.
25. Write a Python program to **merge two sorted lists into a single sorted list without using built-in sorting functions**.